

FINAL PROJECT DSC 335 - AIRBNB PRICE ANALYSIS IN SEATTLE

I. Main effects, first-order, Bayesian regression model

- For the model, we set “flexible” level in the *cancellation_policy* as the baseline
- *Beta-not*: Normal (177, 30²)
- Beds Coefficient: Prior -> Normal(50, 25²)
- Review Scores Rating: Prior -> Normal(2, 1²)
- Cancellation_policy priors:
 - “Strict” level: Normal (40, 5²)
 - “Moderate” level: Normal (20, 5²)
- Sigma: Exp (1/40)

$$\text{Price} = -44.1 + 12.6(\text{cancellation_policy_moderate}) + 24.7(\text{cancellation_policy_strict}) + 43.5(\text{beds}) + 0.855(\text{review_scores_rating})$$

II. Alternative models

- For these following models:
 - The baseline level of “*cancellation_policy*” is adjusted to be “*flexible*”
 - The baseline level of “*room_type*” is adjusted to be “*Shared room*”

a. MODEL 1

- This model includes the first order *review_scores_rating*, and second order of *beds* and *bathrooms*:

$$\text{Price} = 36.5 + 4.25(\text{beds}^2) + 7.98(\text{bathrooms}^2) + 0.596(\text{review_scores_rating})$$

b. MODEL 2

- This model adds the interaction between *beds* and *accommodates*, and the categorical predictor *room_type*:

$$\text{Price} = 6.87 + 2.82(\text{beds}) + 16.9(\text{accommodates}) + 5.85(\text{cancellation_policy_strict}) - 5.40(\text{cancellation_policy_moderate}) + 61(\text{room_typeEntire home/apt}) + 25(\text{room_typePrivate room}) + 1.14(\text{beds} * \text{accommodates})$$

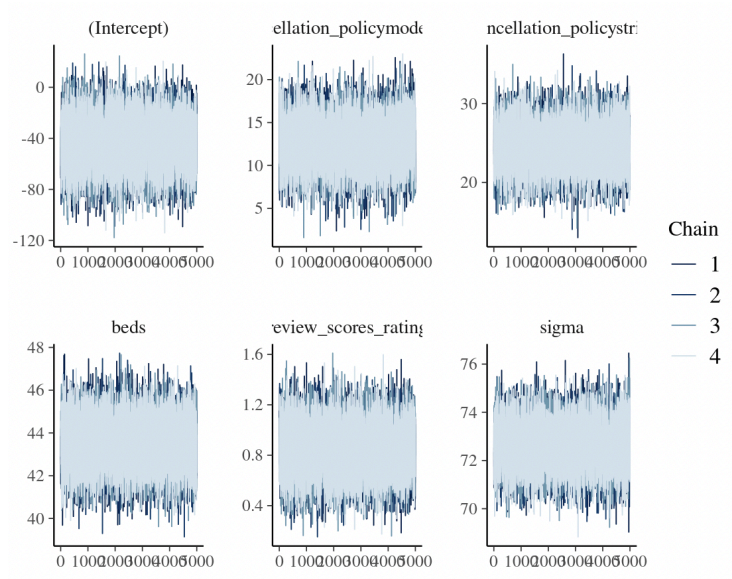
III. Default, weakly informative priors model

$$\text{Price} = -33.7 + 43.6(\text{beds}) + .809(\text{review_scores_rating}) + 1.34(\text{cancellation_policy_moderate}) + 19.2(\text{cancellation_policy_strict})$$

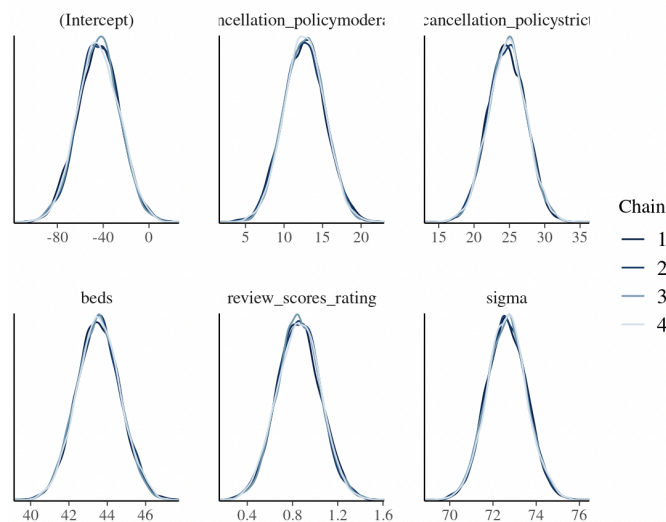
IV. Evaluation and comparison

a. ORIGINAL MODEL

$$\text{Price} = -44.1 + 12.6(\text{cancellation_policy_moderate}) + 24.7(\text{cancellation_policy_strict}) + 43.5(\text{beds}) + 0.855(\text{review_scores_rating})$$



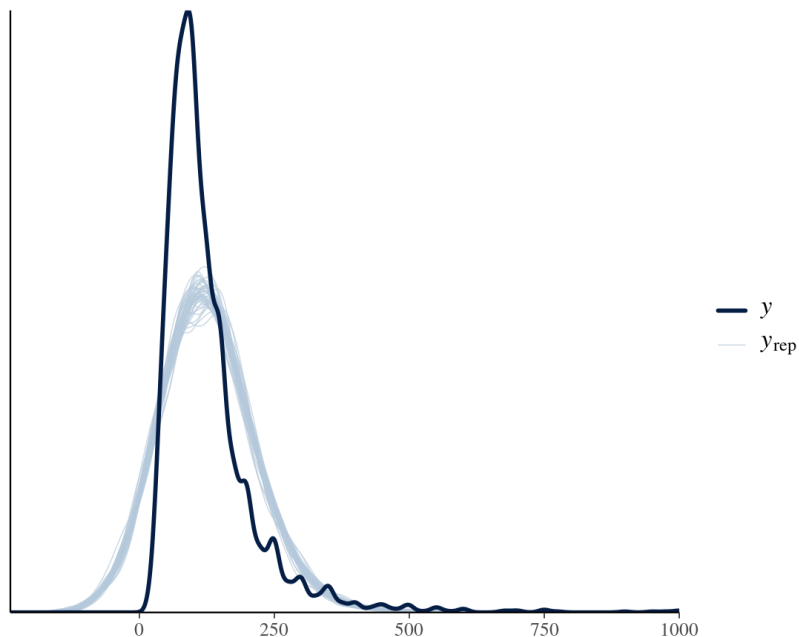
Since our chains are largely overlapping and the chains show reasonable mixing, they appear to have converged



From these density plots, we can tell that our chains are overlapping pretty nicely. An increase of one bed seems like it is predicted to have the largest impact on price (for one unit at least), with a mean of around 44. On the other hand, a one unit increase in review score rating only has a mean of around 0.8, which makes sense considering the range of review scores (0-100).

term	estimate	std.error	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	-44.1	18.4	-74.3	-13.8
2 cancellation_policymoderate	12.6	2.70	8.22	17.0
3 cancellation_policystrict	24.7	2.70	20.2	29.0
4 beds	43.5	1.14	41.7	45.4
5 review_scores_rating	0.855	0.191	0.541	1.17

All of our predictor variables appear to be statistically significant predictors as 0 is not included in any of the credible intervals. Review_scores_rating, however, has the lowest statistical significance, with a 90% posterior credible interval of (\$0.54, \$1.17)



Looking at the center of our simulated data sets, we see that it is roughly the same as the original data, although slightly increased. However, our model clearly does not accurately capture the spread as the spread is consistently larger for every simulated data set.

#Posterior predictive summary:

```
      mae mae_scaled within_50 within_95
1 31.91698  0.4399324 0.6843938 0.9639126
```

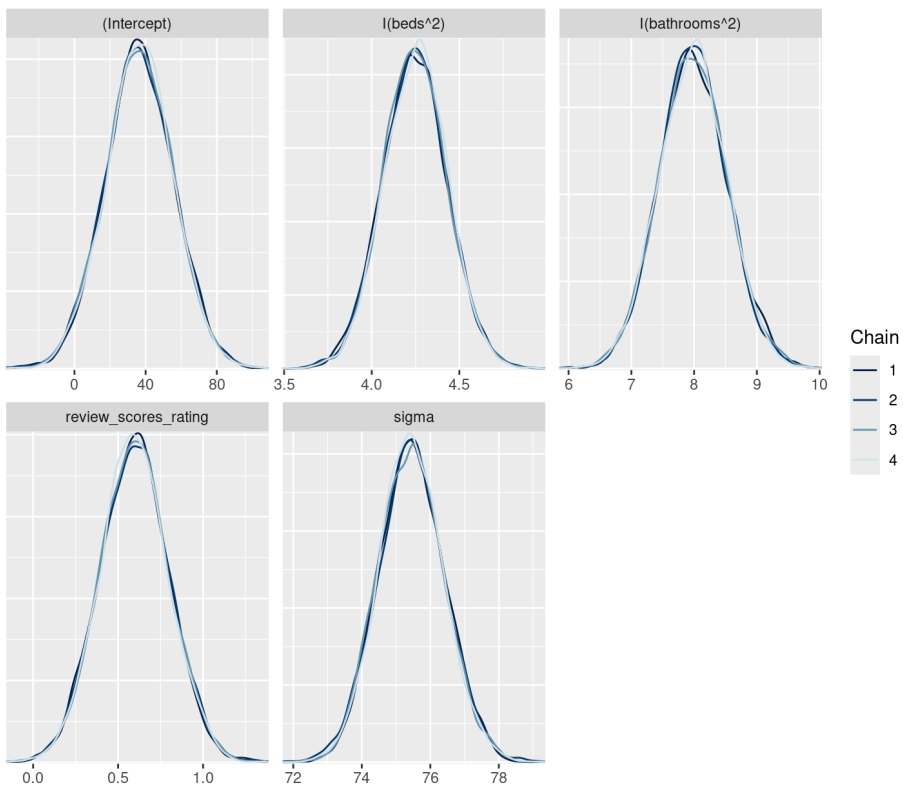
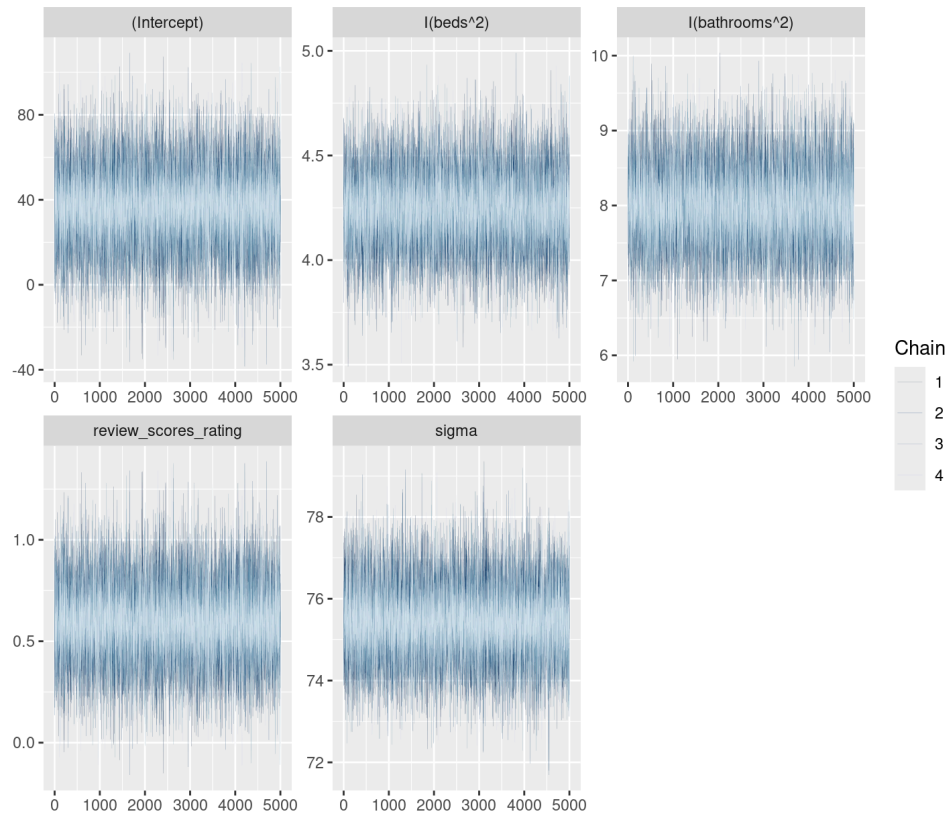
#Cross-validation:

```
      mae mae_scaled within_50 within_95
1 32.43167  0.4465766 0.6837422 0.9635885
```

Our scaled mae numbers are fairly high for both the cross validation posterior predictive check and the regular posterior predictive check.

b. ALTERNATIVE MODEL 1

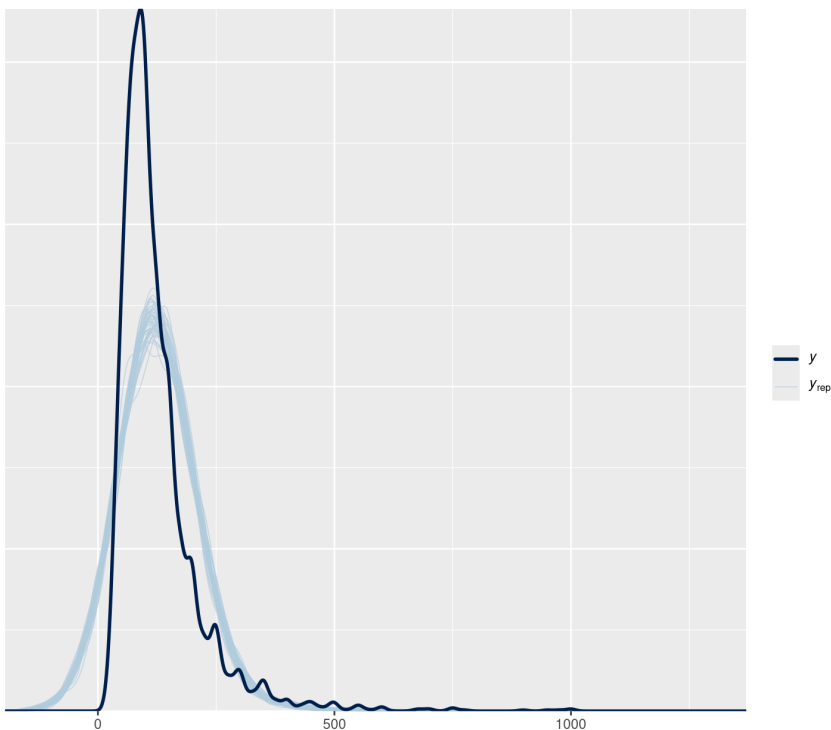
Price = 36.5 + 4.25(beds^2) + 7.98(bathrooms^2) + 0.596(review_scores_rating)



#Trace plot and density plot show stability.

term	estimate	std.error	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	36.5	19.0	4.58	68.2
2 I(beds^2)	4.25	0.181	3.95	4.55
3 I(bathrooms^2)	7.98	0.540	7.10	8.89
4 review_scores_rating	0.596	0.200	0.264	0.934
5 sigma	75.4	0.959	73.9	77.1
6 mean_PPD	127.	1.89	124.	130.

#Posterior credible intervals don't include 0, indicating significance of the variables.



#Pp_check: plotting multiples densities against the original data. The plot shows that while the densities follow a similar pattern and distribution with the original data, there was still a huge difference that poses concern for this model.

mae	mae_scaled	within_50	within_95
34.00752	0.4503214	0.7068693	0.9616967

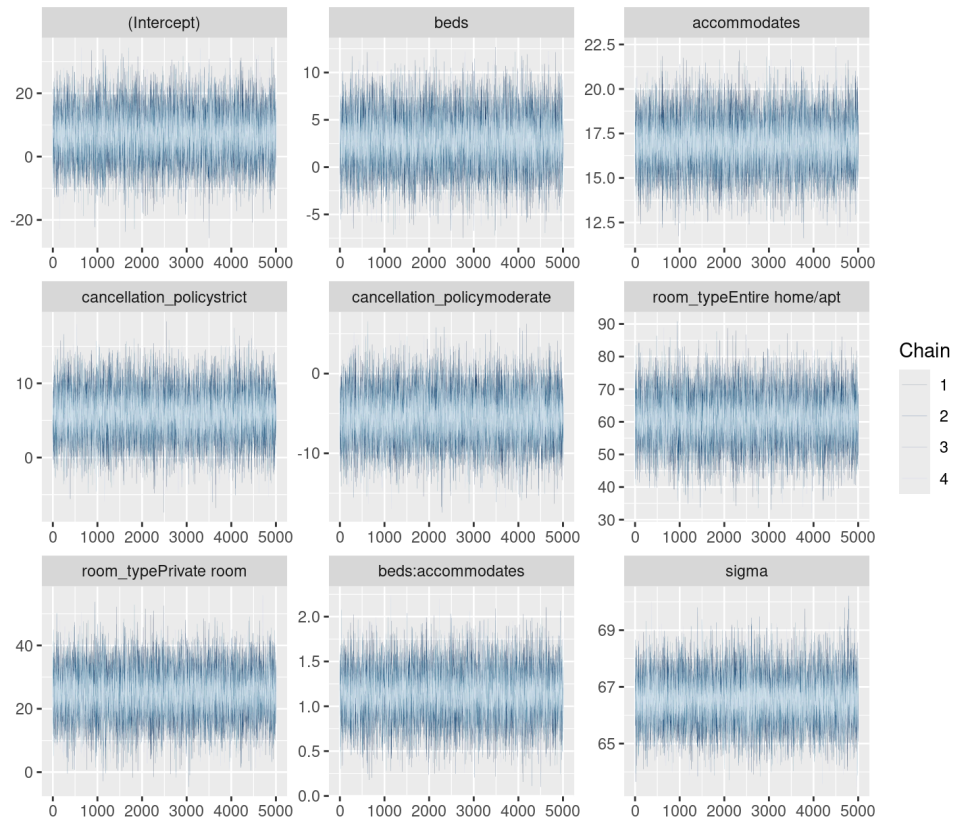
#Posterior predictive summary

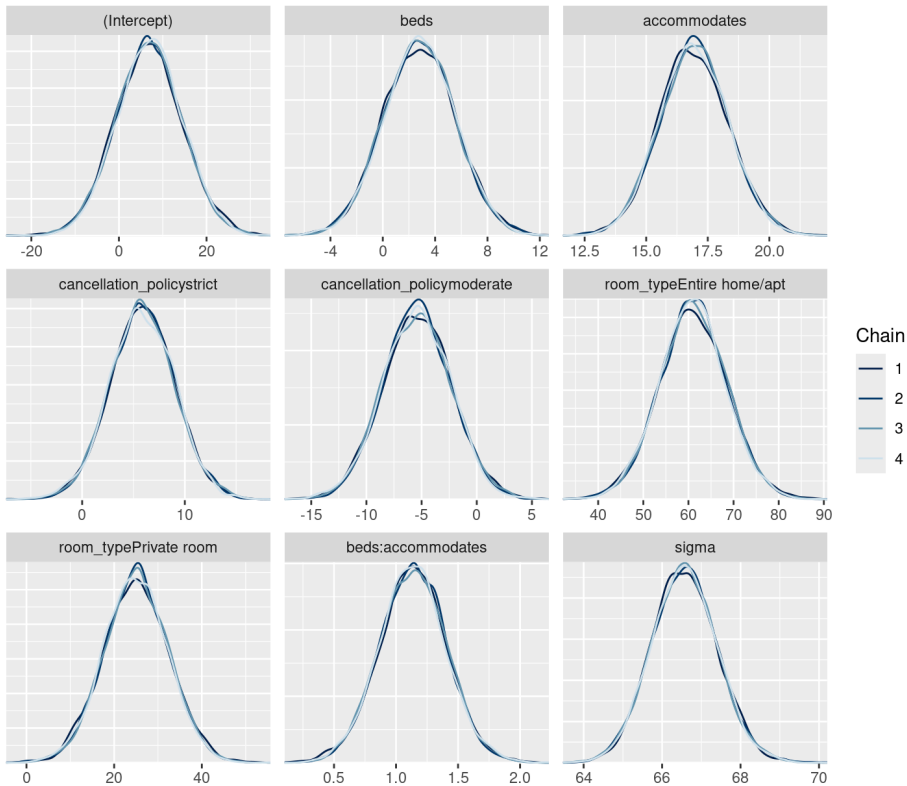
mae	mae_scaled	within_50	within_95
33.84739	0.4495148	0.7084459	0.9610569

#Cross-validation

c. ALTERNATIVE MODEL 2

Price = 6.87 + 2.82(beds) + 16.9(accommodates) + 5.85(cancellation_policy_strict) - 5.40(cancellation_policy_moderate) + 61(room_typeEntire home/apt) + 25(room_typePrivate room) + 1.14(beds*accommodates)

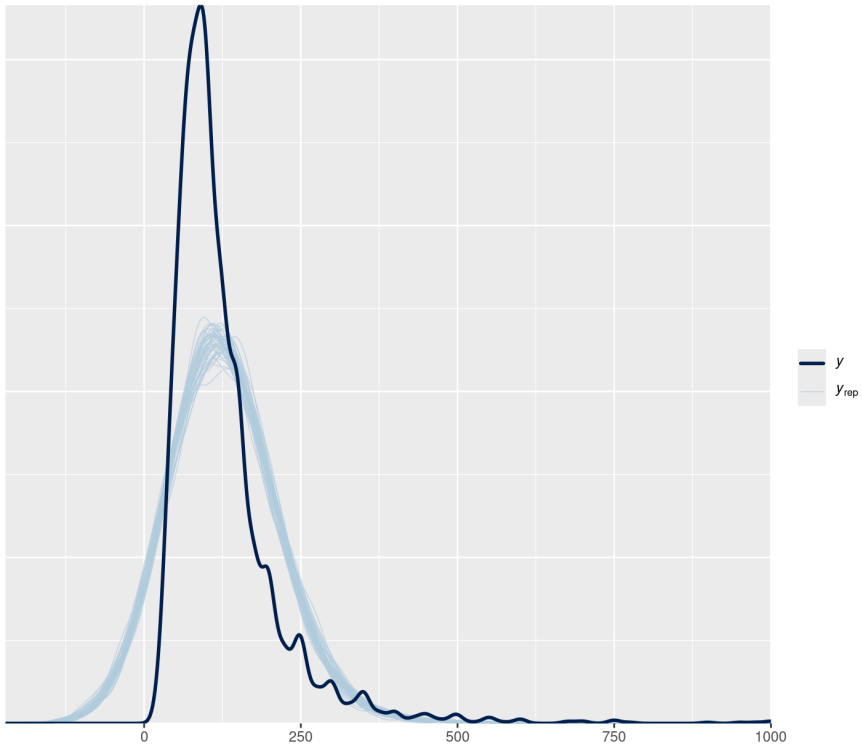




#Trace plot and density plot show stability.

term	estimate	std.error	conf.low	conf.high
<i><chr></i>	<i><dbl></i>	<i><dbl></i>	<i><dbl></i>	<i><dbl></i>
1 (Intercept)	6.87	7.52	-5.62	19.4
2 beds	2.82	2.75	-1.70	7.34
3 accommodates	16.9	1.40	14.7	19.3
4 cancellation_policystrict	5.85	3.14	0.736	11.0
5 cancellation_policymoderate	-5.40	3.13	-10.5	-0.351
6 room_typeEntire home/apt	61.0	7.34	48.6	73.2
7 room_typePrivate room	25.0	7.35	12.8	37.1
8 beds:accommodates	1.14	0.265	0.689	1.58
9 sigma	66.6	0.837	65.2	68.0
10 mean_PPD	127.	1.67	124.	130.

A few of the coefficients' intervals contain 0, indicating that they are not significant in predicting price. Overall, most of the predictors are significant.



#Pp_check: similar to other models', plotting multiples densities against the original data. The plot shows that while the densities follow a similar pattern and distribution with the original data, there was still a huge difference that poses concern for this model.

mae	mae_scaled	within_50	within_95
25.85055	0.3880156	0.7207977	0.9639126

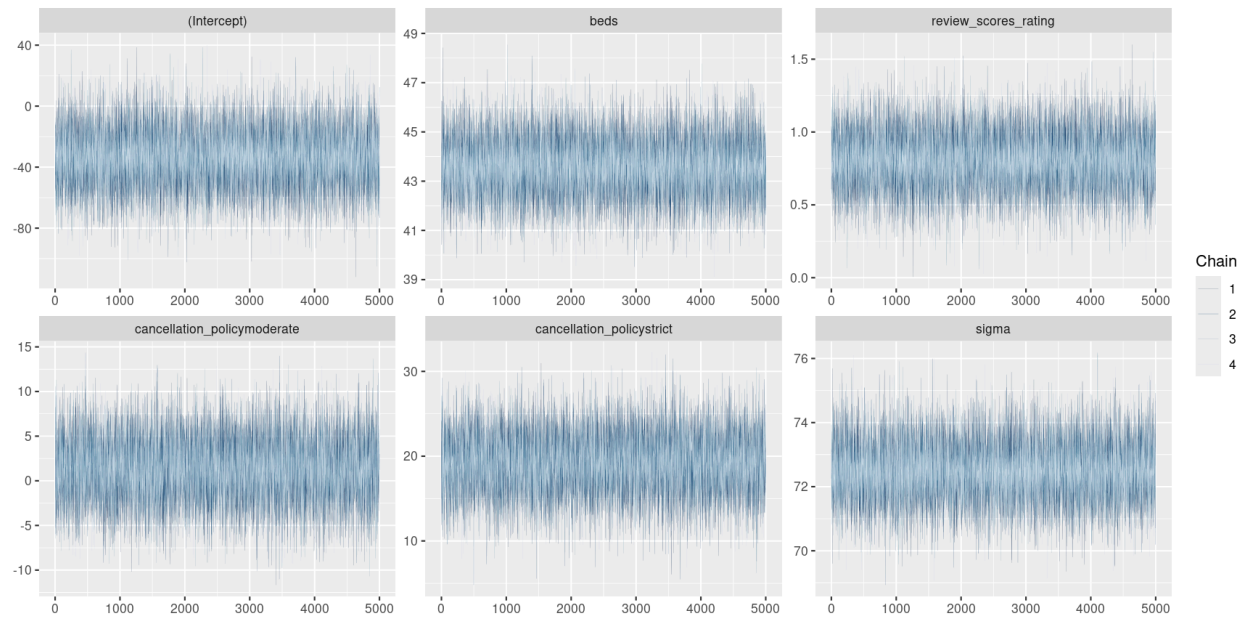
#Posterior predictive summary

mae	mae_scaled	within_50	within_95
26.10535	0.3920606	0.7160589	0.9629546

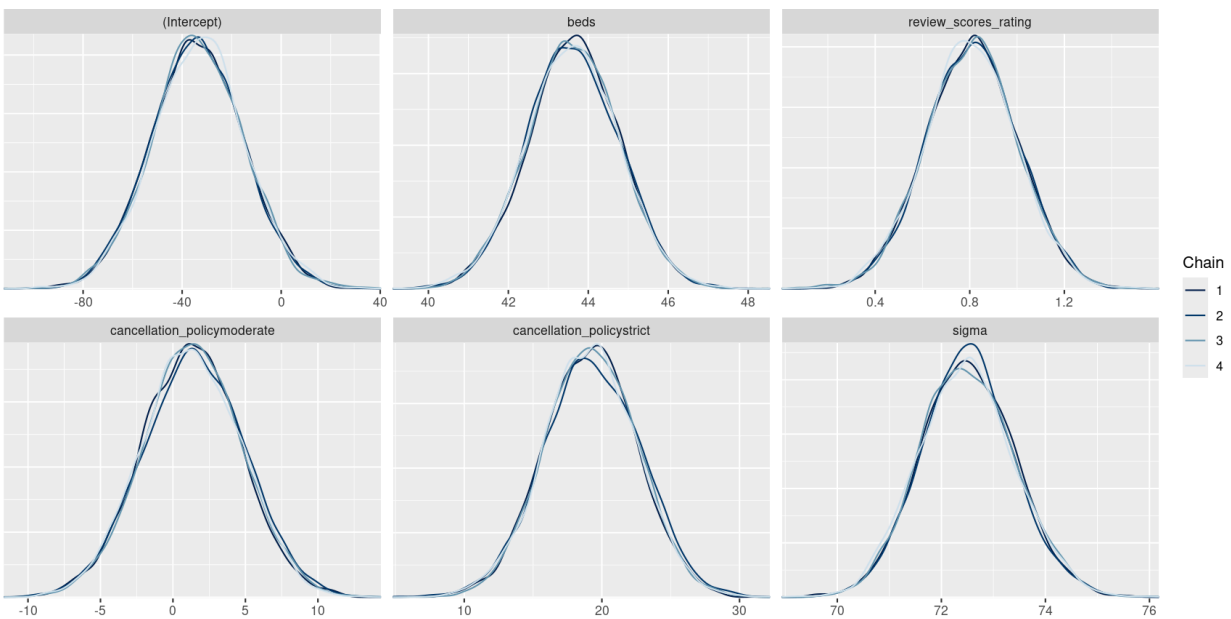
#Cross-validation

d. DEFAULT, WEAKLY INFORMATIVE PRIORS MODEL

$$\text{Price} = -33.7 + 43.6(\text{beds}) + .809(\text{review_scores_rating}) + 1.34(\text{cancellation_policy}\$\text{moderate}) + 19.2(\text{cancellation_policy}\$\text{strict})$$



Trace plots overall are stabilized across chains but the typical intercept distribution explores negative price values for the majority of fit time.



Here we see the same representation as before with the trace plots. This simulation is unsure of how to quantify the effect of cancellation_policy categories.

term	estimate	std.error	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	-33.7	18.4	-64.4	-2.93
2 beds	43.6	1.15	41.7	45.5
3 review_scores_rating	0.809	0.192	0.488	1.13
4 cancellation_policymoderate	1.34	3.44	-4.32	7.00
5 cancellation_policystrict	19.2	3.43	13.6	24.8

The intercept is intuitively way off for this particular price regression problem. Also there appears to be no significant difference between the moderate and flexible levels of cancellation_policy as zero dwells in the posterior interval for the moderate level.

```
mae mae_scaled within_50 within_95
1 31.56082 0.4353746 0.6850269 0.9632795
mae mae_scaled within_50 within_95
1 31.44718 0.4343476 0.6796494 0.9635955
```

Here we see that the MAE is very high with little deviation about the average value. This doesn't change after a 10 fold cv is implemented.

e. CHOICE OF BEST MODEL

- Since our second alternative model has the lowest mae of 26.105 when using cross validation, we chose that model as the best performing one. The mae of the second alternative model is at least 5 fewer than the other three models' mae despite relatively similar within_50 and within_95 values.

V. Summary

term	estimate	std.error	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	6.87	7.52	-5.62	19.4
2 beds	2.82	2.75	-1.70	7.34
3 accommodates	16.9	1.40	14.7	19.3
4 cancellation_policystrict	5.85	3.14	0.736	11.0
5 cancellation_policymoderate	-5.40	3.13	-10.5	-0.351
6 room_typeEntire home/apt	61.0	7.34	48.6	73.2
7 room_typePrivate room	25.0	7.35	12.8	37.1
8 beds:accommodates	1.14	0.265	0.689	1.58
9 sigma	66.6	0.837	65.2	68.0
10 mean_PPD	127.	1.67	124.	130.

Posterior Credible Interval Interpretations

There is a 90% posterior probability that the true increase in price for each additional person accommodated by the airbnb lies between **14.70 and 19.30 dollars**, with all other variables held constant and the number of beds equals 0. This indicates a strong and positive association.

There is a 90% posterior probability that the true effect on price for each additional bed in the airbnb lies between **-1.70 and 7.34 dollars**, with all other variables held constant and number of guests accommodated equal 0. This suggests that beds might not be a significant predictor for price in this model, as 0 occurs inside of the credible interval.

There is a 90% posterior probability that for each additional unit of **accommodates**, the effect of 1 more bed on price increases between **0.689 and 1.58 dollars**, with all other variables held constant.

There is a 90% posterior probability that the true increase in price for having a **strict cancellation policy** (compared to the flexible baseline) with all other variables held constant lies between **0.74 and 11.00 dollars**, suggesting a likely positive association, though there is some uncertainty.

There is a 90% posterior probability that the true decrease in price for having a **moderate cancellation policy** (compared to the flexible baseline) with all other variables held constant, lies between **0.35 and 10.50 dollars** suggesting a likely negative association, though there is some uncertainty.

There is a 90% posterior probability that the true increase in price for having an **“entire home/apartment”** as the room type (compared to the “shared room” baseline) with all other variables held constant, lies between **48.60 and 73.20 dollars** suggesting a strong positive association.

There is a 90% posterior probability that the true increase in price for having a **“Private Room”** as the room type (compared to the “shared room” baseline) with all other variables held constant, lies between 12.80 and 37.10 dollars suggesting a strong positive association.

There is a 90% posterior probability that the true increase in price caused by a one unit increase in the **beds-to-accommodate ratio**, with all other variables held constant, is between **0.69 and 1.58 dollars**. This suggests a positive association with some uncertainty.

VI. Posterior predictive analysis

$$\text{Price} = 6.87 + 2.82(\text{beds}) + 16.9(\text{accommodates}) + 5.85(\text{cancellation_policy_strict}) - 5.40(\text{cancellation_policy_moderate}) + 61(\text{room_typeEntire home/apt}) + 25(\text{room_typePrivate room}) + 1.14(\text{beds} * \text{accommodates})$$

We used the following set of covariates for our variable of interest:

- Cancellation_policy = "Strict"
- Beds = 2
- Accommodates = 4
- Room_type = Entire home/apt

The 90% posterior predictive interval for this set of covariates is **(47.13, 265.86)**. There is a 90% posterior probability that the price of a listing that has strict cancellation policy, is the entire house/apt, 2 beds and accommodates 4 people is between **47.13 and 265.86 dollars**, which, due to being such a wide interval, is not very useful for predicting.